Hamza Khan

Senior Backend Engineer, Architect & Tech Lead

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SUMMARY

6+ years designing, building, and operating distributed systems and cloud-native backend platforms — from high-availability APIs to AI-powered clinical workflows. Architected and led a HIPAA-compliant telehealth platform from the ground up; experienced in service boundaries, async pipelines, AWS infrastructure, observability and mentoring engineers.

EXPERIENCE

Ignis Health — Senior Software Engineer · Backend / Architect / Tech Lead **Dec 2022 – Present**

USA (Remote)

- Architected and led development of Axis-360, a HIPAA-compliant telehealth platform built from the ground up — covering clinical workflows, scheduling, billing, and AI-assisted documentation.
- Owned backend service architecture and boundaries; designed scalable APIs and data models using Django REST Framework and PostgreSQL for thousands of daily active users.
- Built fault-tolerant async pipelines using Celery, Redis, and cloud-managed messaging to process high volumes of clinical and operational events with guaranteed delivery.
- Improved system performance through query optimization, caching strategies, and indexing — reducing P95 API latency by ~50%.
- Led cloud infrastructure migration to AWS (EKS, RDS, S3, CloudFront), improving availability, deployment safety, and operational visibility.
- Integrated LLM-powered workflows to automate clinical documentation and summarization, reducing manual effort for clinicians.
- Implemented CI/CD pipelines with automated testing and blue/green deployments, enabling safe and frequent releases.
- Established observability standards (Prometheus, Grafana, centralized logging) to monitor system health and background job throughput; mentored junior engineers and drove architectural decisions.

InvoZone — Software Engineer · Backend **Dec 2020 – Dec 2022**

Pakistan

- Designed and shipped backend services for multiple cloud-native client products using Django and RESTful APIs, supporting tens of thousands of daily active users.
- Built event-driven workflows using async task processing and messaging systems to decouple request handling from long-running background jobs.
- Deployed serverless workloads on AWS Lambda and GCP Cloud Run for background processing, scheduled tasks, and API extensions.
- Containerized microservices with Docker and Kubernetes, enabling horizontal scaling and consistent environments across staging and production.
- Contributed to blockchain-integrated backend systems — implemented secure smart contract interactions and off-chain processing pipelines.
- Designed CI/CD pipelines with automated testing, reducing manual release intervention by ~80%; integrated Stripe and Auth0 for multi-tenant SaaS billing and authentication.
- Built data ingestion and transformation pipelines with a focus on reliability, idempotency, and observability.

Taleemabad — Associate Software Engineer **Jul 2019 – Dec 2020**
Pakistan

- Developed and maintained backend services for a national-scale LMS serving 100K+ active students across Pakistan.
- Built async background workflows using task queues and Redis for notifications, content sync, and analytics processing.
- Implemented real-time features using WebSockets and Redis pub/sub for live classroom interactions.
- Optimized PostgreSQL queries and data models for high-traffic endpoints; integrated search and indexing to improve content discovery across large educational datasets.
- Supported production deployments and monitoring of backend systems under real load.

OPEN SOURCE

Volta Engine — serverless microVM runtime **github.com/khanashes/volta**
Firecracker/KVM microVMs with snapshot/restore and userfaultfd lazy memory restore; resumes VMs before memory is fully copied. Targets sub-100ms cold starts, sub-200ms lazy restore. Python SDK + voltad daemon.

FlowMesh — distributed event streaming **github.com/khanashes/flowmesh**
Append-only partitioned logs with consumer groups, replay semantics and backpressure; built to explore ordering, durability and fault-tolerance trade-offs.

SKILLS

Languages	Python, SQL, JavaScript, TypeScript, Bash
Backend	Django, Django REST Framework, FastAPI, Celery
Datastores	PostgreSQL, Redis, MongoDB, Elasticsearch
Cloud & infra	AWS (EC2, EKS, ECS, Lambda, RDS, S3, CloudFront), GCP (Cloud Run, Pub/Sub), Docker, Kubernetes, Helm, Terraform
DevOps & observability	GitHub Actions, Prometheus, Grafana, CloudWatch, blue/green deployments, GitOps
Security	OAuth2 / JWT, RBAC, HIPAA compliance, encryption at rest & in transit

RECOGNITION & EDUCATION

- IEEE Xtreme Programming Competition — International, Top 3 finish
- Growth Driver of the Year — InvoZone internal recognition award
- AWS Certified Cloud Practitioner (in progress) · IBM Python for Data Science · Advanced Django (Udemy)
- B.S. in Software Engineering — COMSATS Institute of Information and Technology, Islamabad (2016 – 2020)